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Organic Electrochemical Transistors Based on Blend Films with Thermoresponsive Polymer

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Organic electrochemical transistors (OECTs) are biocompatible devices with significant potential for biosensing. Functionalizing the channel layers is essential for improving the selectivity and sensitivity of OECT-based biosensors. A straightforward one-step fabrication method for these functionalized channel layers can simplify the production process for these devices. This study developed OECT devices based on a polymer blend of poly(3,4-ethylenedioxythiophene)-poly(styrenesulfonate) (PEDOT:PSS) and poly(*N*-isopropylacrylamide) (PNIPAM) that respond to temperature changes. Structural analyses of the blended films showed that hole transport through PEDOT is maintained even after blending, and the PNIPAM is segregated at the surface. To overcome the large chain conformational change that occurs with temperature changes, a flexible poly(ethylene glycol) diglycidyl ether (PEGDE) crosslinker is used in addition to the conventional crosslinker, (3-glycidyloxypropyl)trimethoxysilane (GOPS). As a result, the PEGDE + GOPS binary crosslinker system exhibited reversible responses to temperature cycling. These results highlight two key considerations when designing a functional mixed-conductor film based on a polymer blend system: (1) vertical phase separation and (2) proper crosslinker selection.

1. Introduction

Organic mixed ionic–electronic conductors are a vital class of functional materials for soft electronics and bioelectronic applications.[1–4] These materials transport ions and electrons/holes through hydrophilic ion transport sites and π-conjugated electron/hole transport sites, respectively. Thus, the fine-tuning of the chemical structures at the molecular level and film structures at the nano- and submicrometer levels can achieve high conductivity and coupled transport properties.[5–10] Organic electrochemical transistors (OECTs) are one application of these materials, and extensive research has been carried out, from fundamental studies on their operating mechanisms to the development of their practical applications.[9,11–16] These transistors are three-terminal switching devices that resemble organic field-effect transistors (OFETs). However, in contrast to conventional OFET devices, the channel layer of OECT devices is immersed in an aqueous electrolyte, injecting ions from the electrolyte into the bulk channel layer. This unique feature provides biocompatibility, enabling applications such as in vitro biosignals, biomarkers, and body surface monitoring, as well as the potential for implantation.[17–24] For bioelectronic applications, the functionalization of devices is as critical for the development of in vivo probes based on multichannel OECT arrays with enzymes,[18,20,21] ionophores,[19,24] and molecularly imprinted membranes.[22,23] Such OECT arrays will realize parallel and simultaneous monitoring of multiple biomarkers and molecules.

Preparing functionalized channel layers is key to realizing bio-probes based on OECT arrays. One method for obtaining functionalized OECT devices is to form multiple layers of channels and functional layers separately using overcoating processes. This method ensures good device performance and stability but requires multiple sequential coating processes. On the other hand, the single-layer approach, based on blend films of mixed conductors and functional materials, is a straightforward method for preparing bioelectronic devices in a single coating process. For instance, blending films with polysaccharides such as xanthan gum[25] or biomaterials such as *Aloe vera*[26] enhances the biocompatibility of PEDOT-based bioelectrodes. In this

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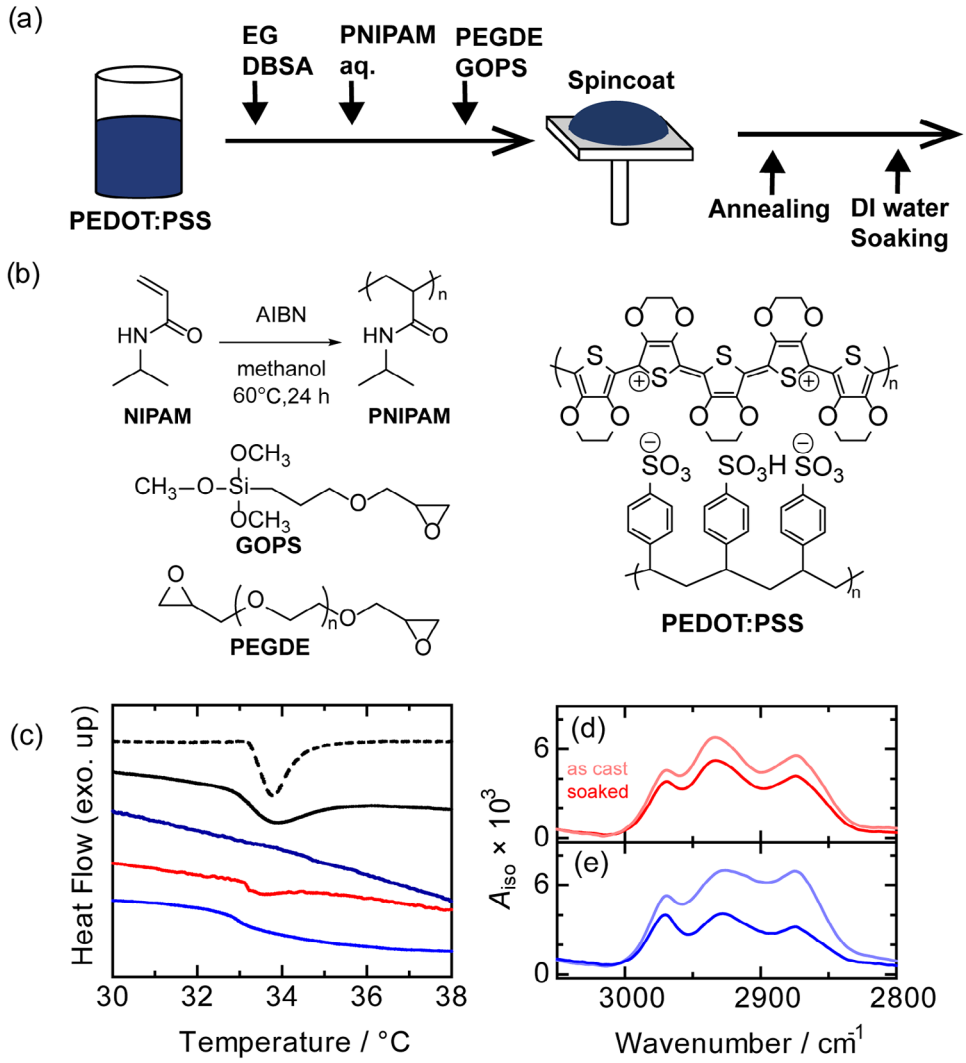


Figure 1. a) Schematic of film preparation, b) chemical structures of materials, c) DSC curves of PNIPAM aqueous solution (black, broken line), PNIPAM film crosslinked by GOPS (black, solid line), and 1:2 blend films of PEDOT:PSS and PNIPAM crosslinked by GOPS (red) and PEGDE + GOPS (blue). d,e) Orientation-free A_{iso} spectra of PEDOT:PSS + PNIPAM (1:2) blend films before (pale curves) and after soaking in DI water (dark curves) crosslinked with GOPS (red) and PEGDE + GOPS (blue).

approach, the most significant challenge is to secure the electronic conductive path in the channel region and the stability of the film under aqueous electrolyte soaking, voltage application, and environmental changes, such as temperature. Therefore, investigating and controlling phase-separated structures is essential to obtain stable and reliable high-performance OECT devices.

In this paper, we present an approach to functionalize OECT devices with a thermoresponsive polymer, poly(*N*-isopropylacrylamide) (PNIPAM, **Figure 1**), as a case study for the development of the fabrication technique of a functionalized channel in an OECT based on poly(3,4-ethylenedioxythiophene)-poly(styrenesulfonate) (PEDOT: PSS), which has been widely used for bioelectronics due to its excellent biocompatibility and electronic property. Here, we employed PNIPAM, which has a lower critical solution temperature (LCST) of ≈32 °C owing to coil-to-globule transition coupled with dehydration of hydrophilic moieties.[27–34] This transition enables temperature sensing applications for the temperature compensation of each OECT sensor on a bioprobe with multiple transistor arrays for accurate biosensing. For the blend film of PNIPAM and mixed conductors, blend films should be sufficiently flexible to withstand volume changes caused by the coil-to-globule transition of polymer chains. Lopez-Larrea *et al.* recently reported PEDOT:PSS + PNIPAM hydrogel based on the crosslinked PIPAM polymer network prepared by photopolymerization that realizes a temperature-responsive OECT when used as a gate electrode.[35] On the other hand, PNIPAM functionalization of the channel layer has not been achieved so far. In this study, we attempt to create chemically crosslinked and surface-anchored PEDOT:PSS + PNIPAM channel layers that can be fabricated by combining wet coating and thermal annealing, considering the process compatibility for neuroprobe fabrication. We used a flexible crosslinker to overcome this challenge in a proof-of-concept study: a PEG-based crosslinker, poly(ethylene glycol) diglycidyl ether (PEGDE), was

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added to the conventional rigid crosslinker, (3-glycidyloxypropyl) trimethoxysilane (GOPS).[36,37] In this mixture, GOPS serves as both a surface anchoring agent to the substrate and a crosslinker of the film, while PEGDE is a flexible crosslinker. This is based on the chemical reactions of epoxy groups for polymers and silanol groups for hydroxide groups on the surface of the glass substrate. Previous studies have incorporated PEG-based crosslinkers into mixed conductors to improve flexibility[38–40] and PEGDE as a flexible crosslinker for PEDOT: PSS films.[41–43] Structural analysis indicated that PNIPAM segregated at the surface while adding PEGDE enhanced the crystallinity of PEDOT. The increased crystallinity from PEGDE contributes to improved device performance by enhancing electronic mobility. Furthermore, the blend film containing PEGDE and GOPS demonstrated a reversible response to temperature cycling. These results suggest a rational design approach for blend film-based organic electrochemical transistors (OECT) intended for biosensing applications in biocompatible environments.

2. Result and Discussion

The stability of neat PEDOT:PSS films were tested using various molar ratios of PEGDE and GOPS crosslinkers. GOPS and PEGDE were selected as hard and flexible crosslinkers, respectively. All films containing GOPS remained adhered to the substrate even after soaking in DI water for two days, whereas the film crosslinked only by PEGDE peeled off from the substrate, although the film shape was maintained (Figure S3, Supporting Information), indicating that only GOPS was responsible for adhesion to the glass substrate. From this result, PEGDE + GOPS (8:2) was the optimal binary crosslinker system, as the PEGDE composition was the highest among the films that did not peel off from the glass substrate. The UV–vis spectra support the stability of the film. The peaks at ≈225 nm, which was attributed to the aromatic rings of PSS,[44–47] show only a slight change upon soaking in DI water, indicating that PEDOT: PSS remained intact even after immersion in DI water (Figures S4 and S5, Supporting Information). The thermal responses of the PNIPAM aqueous solution and the crosslinked PNIPAM film were evaluated using DSC (Figure 1c). Endothermic peaks were observed at ≈33 °C for PNIPAM crosslinked by GOPS (33.5 °C), PEDOT:PSS + PNIPAM blend films crosslinked by GOPS (33.1 °C), and PEDOT:PSS + PNIPAM blend films crosslinked by PEGDE + GOPS (8:2) (32.7 °C), which is close to the critical temperature of PNIPAM in aqueous solution (32.9 °C).[27–31] This indicates that the crosslinking reaction did not hinder the coil–globule transition of the PNIPAM chains.

Figure 1d,e shows the orientation-free spectra of the blended films before and after immersion in DI water to elucidate the elution of PSS from the film.[49] These spectra were obtained from infrared IR-pMAIRS to estimate the isotropic absorption A_{iso} for quantitative analysis of each component free from the molecular orientation that influences the peak intensity via the surface selection rules, $A_{iso} = \frac{1}{3}(2A_{IP} + A_{OP})$.

Here, A_{IP} and A_{OP} denote the in-plane and out-of-plane absorption spectra, respectively.[50–52] A_{iso} is free from optical anisotropy that affects the absorbance of simple transmission IR spectra. The band at ≈2970 cm^{-1} can be assigned to the CH_3 antisymmetric stretching vibration ($\nu_{as,CH3}$),[53,54] which originates only from PNIPAM and decreased by 18% after DI-water immersion for blend films with GOPS. This indicates that 82% of PNIPAM remains even after immersion in DI water. In contrast, the CH_2 symmetric ($\nu_{s,CH2}$) and antisymmetric stretching vibrations ($\nu_{as,CH2}$) (≈2870 and 2930 cm^{-1}) showed larger decreases compared to the $\nu_{as,CH3}$ peaks by 34 and 30%, respectively. These peaks were derived from both PNIPAM and PSS, suggesting the elution of PSS into DI water, as previously reported.[55–58] However, the as-cast blend films with PEGDE + GOPS crosslinkers showed a larger absorbance of $\nu_{as,CH2,}$ and $\nu_{s,CH2}$ due to the absorption by PEGDE. These peaks decreased after the water soaking (reduced by 68 ($\nu_{as,CH3}$), 75 ($\nu_{as,CH2}$), and 80% ($\nu_{s,CH2}$) of the as-cast films) because of the elution of PEGDE, likely because of the lack of alkoxy silane (-SiOR) groups that react with hydroxyl groups on the substrate resulting in self-condensation reactions.[59] The Raman spectra (Figure S6, Supporting Information) show a peak at ≈1430 cm^{-1}, corresponding to the C=C stretching vibration of the PEDOT thiophene rings.[60–62] This peak is known to be sensitive to doping levels and can shift by up to 40 cm^{-1} upon de-doping.[62] However, in these samples, the peak shifts by less than 3 cm^{-1}, suggesting that blending with PNIPAM or crosslinking with PEGDE does not significantly alter the doping levels. These results indicate that PEGDE acts as a crosslinking agent and has good adhesion when combined with GOPS.

Next, we evaluated the nanostructures of the blended films using GIXD measurements. **Figures** 2a,b show the GIXD diffraction patterns of the blend films with GOPS and PEGDE + GOPS crosslinkers, respectively. These patterns have broad halos centered at 6 and 14 nm^{-1}, which correspond to the lamellar stacking of the PEDOT chains (h00) and the PSS amorphous regions, respectively.[63–69] In addition, there is a reflection corresponding to ≈18 nm^{-1}, which is related to the lamellar stacking of the PEDOT chains (h00) and PSS amorphous regions, respectively.[63–69] Further, a peak at ≈18 nm^{-1} was observed in the out-of-plane direction, which corresponds to the (020) plane related to the π stacking of the crystallized PEDOT (Figure S7, Supporting Information). Figures 2c,d show the radial integrals from the in-plane (5°–15°) to the out-of-plane (75°–85°) direction of the π-π stacking peaks, which was used to investigate the molecular orientations. As shown, the scattering intensity is stronger in the out-of-plane direction, indicating the face-on orientation of the π stacking of PEDOT chains. The intensity and line width of this region were more pronounced in the blend film with PEGDE + GOPS crosslinkers than in the film with the GOPS crosslinker. This indicates that the binary crosslinker increased the crystallinity of PEDOT in the films. Notably, the weaker diffraction from the lamellar stacking of PEDOT (h00) in the PEGDE + GOPS film is consistent with a previous report on the inverse correlation between the area of this peak and the crystallinity of π stacking for PEDOT:PSS films with different amounts of added ethylene glycol.[63] Figure 2e shows the linewidth estimated by fitting two Gaussian functions (Figure S8, Supporting Information). The half-width at half-maximum (HWHM) values of the films with PEGDE + GOPS crosslinkers were smaller than those with GOPS crosslinkers. For instance, HWHM of PEDOT:PSS + PNIPAM films reduces from 1.596 to 1.361 nm^{-1} on the introduction of PEGDE. This result indicates that PEGDE increased the size of the crystallites of PEDOT. Polar additives improve the

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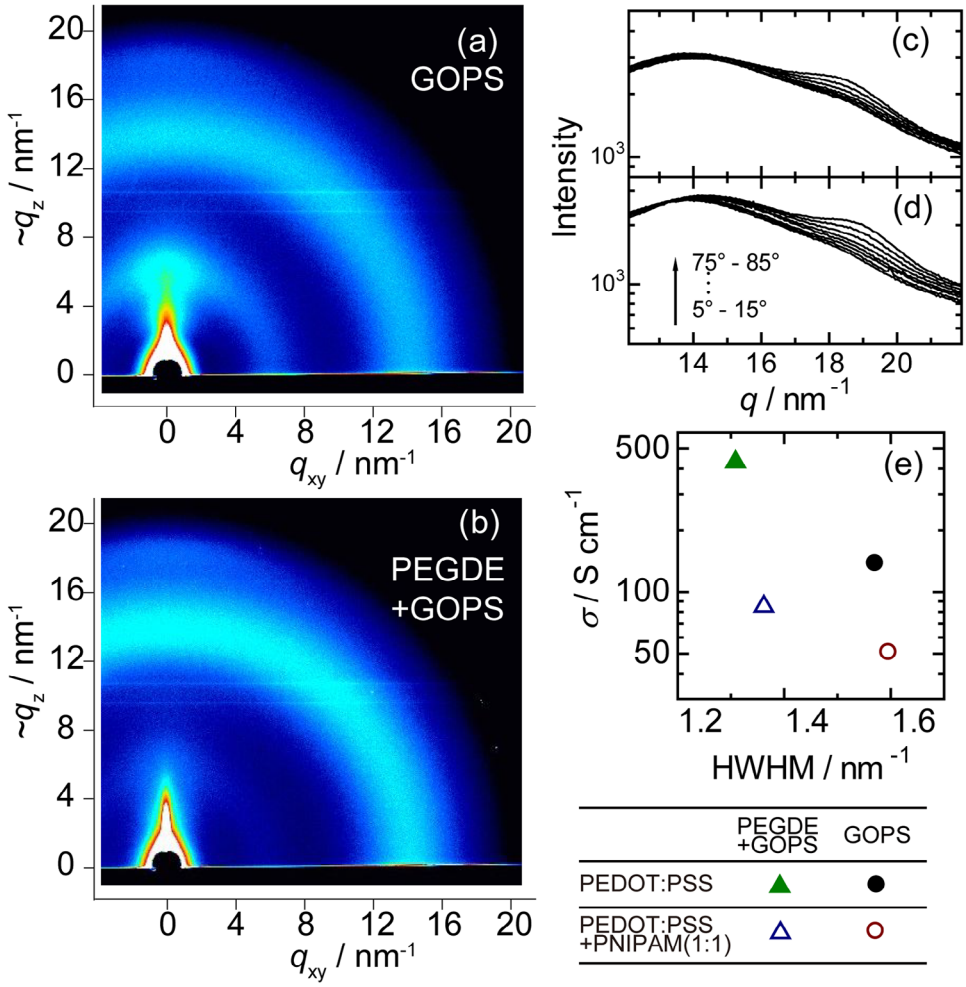


Figure 2. GIXD analysis of blend films: a,b) GIXD patterns of PEDOT:PSS + PNIPAM(1:1) blend films crosslinked with (a) GOPS and (b) PEGDE + GOPS. c,d) Radial integration of intensity from 5° to 85° for (a) and (b). e) Plot of linewidth at ≈18 nm^{-1} and conductivity of films.

electrical conductivity of PEDOT:PSS films.[55–58] For example, Mengistie *et al*. reported that adding EG or PEG to PEDOT:PSS solutions increases the conductivity depending on the molecular weight or chain length of the additive.[70] The addition of PNIPAM caused a slight change in the half width half maximum (HWHM) of each crosslinker system, indicating that PNIPAM decreases the crystallinity of PEDOT. The conductivity measured using the four-point probe method correlated with the crystallinity of PEDOT; the addition of insulating PNIPAM reduced the conductivity. These results confirmed that blending with PNIPAM slightly decreased the crystallinity; however, the PEGDE crosslinker enhanced the crystallinity more significantly than the addition of PNIPAM.

In addition, XPS measurements were performed to examine the distribution of each component in the depth direction. **Figures** 3a-d show the depth-resolved XPS spectra of the films crosslinked with the PEGDE + GOPS and GOPS crosslinkers during Ar^+ cluster ion etching. Notably, Ar^+ cluster ion etching did not damage the sample surface, as previously reported,[71,72] but monoatomic Ar^+ ions degraded the surface. The N1s region contained a peak corresponding to the secondary amide of PNIPAM. The spectral shape of N1s is maintained and the intensity decreases on the supporting time. For a more detailed analysis, Figure 3e,f show the depth profiles. In both blend films with GOPS and PEGDE + GOPS crosslinkers, the atomic concentration of nitrogen was higher in the surface region than in the bulk film. This result indicates that the structure comprised segregated PNIPAM on the surface because nitrogen atoms were contained only in the amide groups of PNIPAM in this system. Such surface segregation of a lower-surface-free-energy component is generally driven by the difference in surface energy to minimize the interfacial free energy.[73–75] For the GOPS and PEGDE + GOPS crosslinkers, the thicknesses of the segregated layers were 9% and 6% of the total film thickness, respectively. The smaller percentage of the segregated phase in the blend film with PEGDE + GOPS crosslinkers is likely due to the high elution of PNIPAM in the PEGDE + GOPS system. This is supported by the fact that PNIPAM dissolved completely even after the crosslinking reaction using PEGDE and immersion in DI water for two days, whereas the PNIPAM film remained crosslinked with GOPS. The S2p region changed with respect to the sputtering time, especially for the PEGDE + GOPS system: the signal

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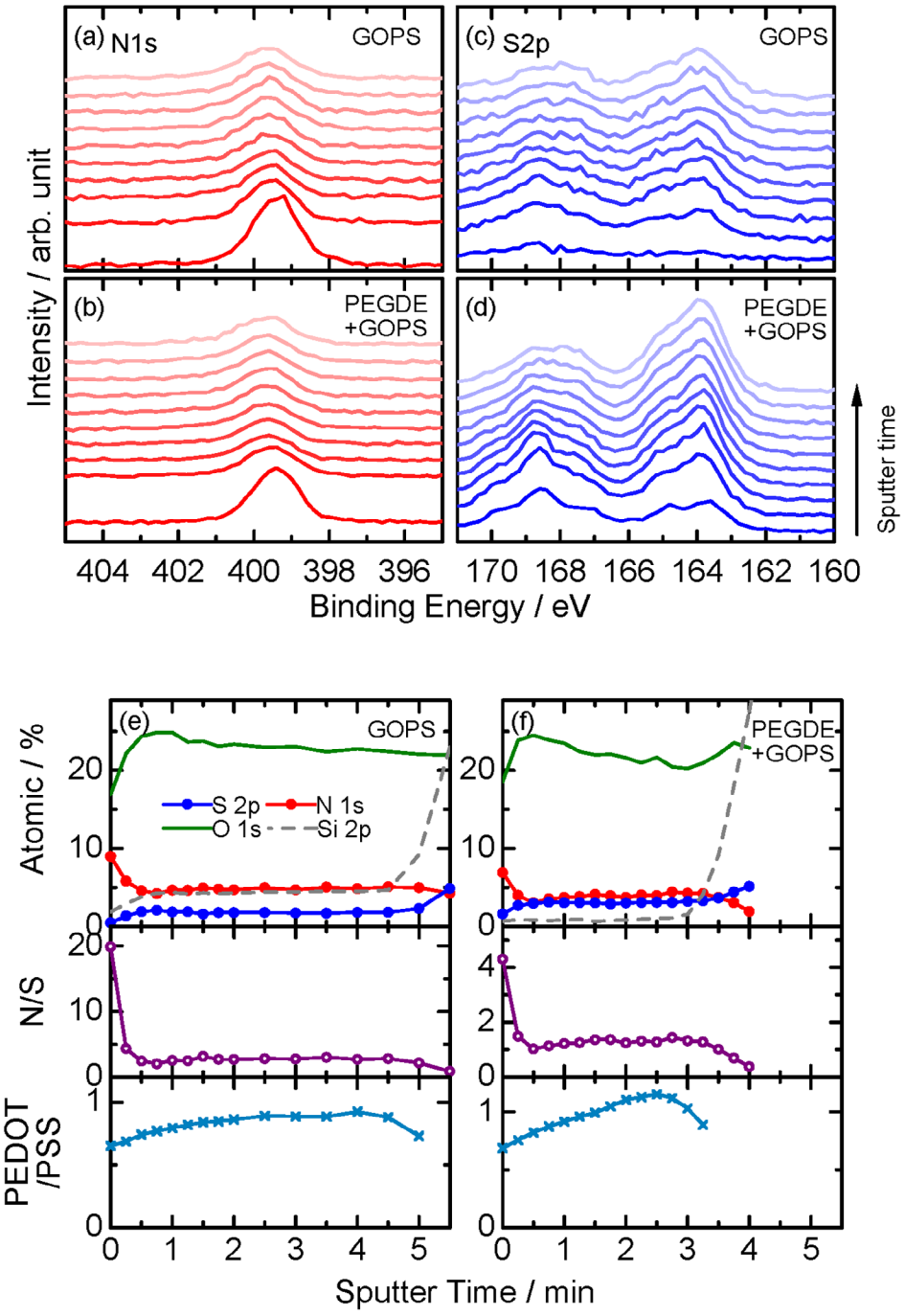


Figure 3. Depth profile of XPS: a,b) N1s and c,d) S2p XPS spectra of 1:2 blend films crosslinked with (a,c) GOPS and (b,d) PEGDE + GOPS (8:2) measured after Ar cluster ion sputtering (0.5 keV). e,f) analyzed XPS depth profiles, ratio of nitrogen and sulfur (N/S), and PEDOT to PSS ratioduring sputtering.

from PEDOT(164 eV) became stronger than PSS (168 eV) with the increase in sputtering time. This indicates the segregation of PSS on the surface, as previously reported.[36,66] The area of these components was estimated and summarized as PEDOT-to-PSS ratio, indicating the PEDOT-rich region at the bottom of the film. It is noteworthy that the PEDOT content in the bottom layer is higher in the presence of PEGDE, which agrees with the high crystallinity of PEDOT in the mixed crosslinker system. The high PEDOT content is beneficial to the hole transport in the channel layer.

To elucidate the surface composition of the blend films crosslinked with PEGDE + GOPS **Figure 4** shows images obtained using photoinduced force microscopy (PiFM). PiFM is an AFM-IR method for mapping contrast from chemical

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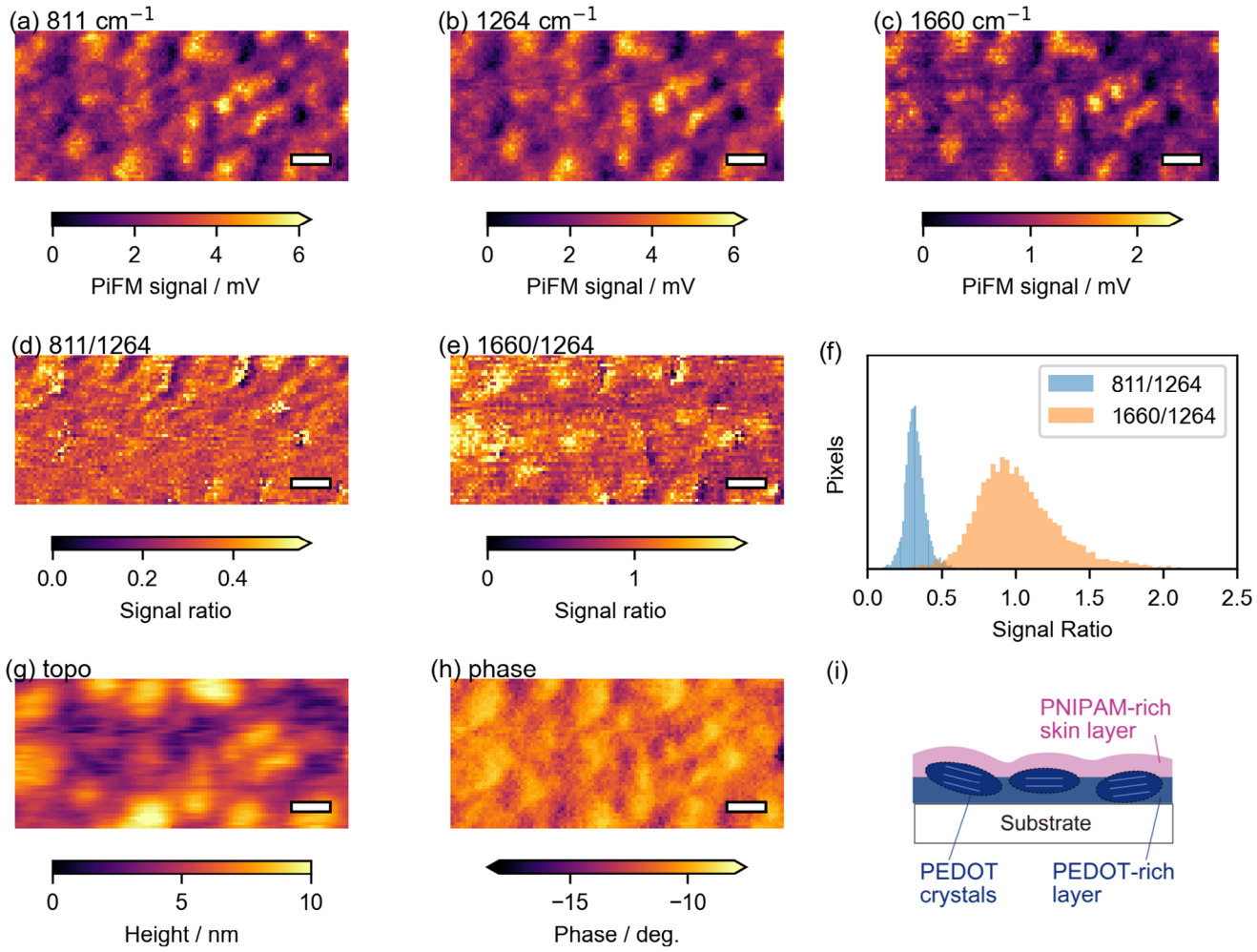


Figure 4. PiFM images of blend films: a–c) PiFM image of the 1:1 blend film crosslinked with PEGDE + GOPS at 811 cm^{-1} (PSS), 1264 cm^{-1} (PEDOT), and 1660 cm^{-1} (PNIPAM). d,e) Processed images of a–c by dividing two images. g,h) Topography and phase images of the corresponding region of the PiFM observations. All scale bars correspond to 100 nm. i) Schematic illustration of the film structure elucidated from analysis.

vibrations at nanometer resolutions. Figures 4a–c were obtained at 811, 1264, and 1660 cm^{-1}, corresponding to the vibrations of PSS (γ(C-H)), PEDOT (ν(Cα- Cα')), and PNIPAM (ν(C═O)), respectively. These images show features corresponding to the surface topography and phases shown in Figure 4g,h, respectively. To confirm the uniformity, we calculated the signal ratio based on the PEDOT signal, as shown in Figure 4d,e. These images show an unimodal distribution of signal ratios (Figure 4f), which implies that the PNIPAM and PSS components were distributed uniformly in the plane direction. The topography of the blend films with GOPS and PEGDE + GOPS crosslinkers were similar, having root-mean-squared (RMS) roughness values of 5.6 and 4.5 nm, respectively. We confirmed that the roughness of neat PEDOT:PSS films (1.6 and 2.0 nm with GOPS and PEGDE + GOPS crosslinkers, respectively) was smaller than that for blend films, and the ratio between RMS roughness and the thickness was more significant in blend films with a large PNIPAM content (Table S1, Supporting Information). Based on the results shown in Figures 3 and 4, we conclude that the film has a segregated layer of PNIPAM on top, followed by a PSS-rich interlayer and a PEDOT-rich bottom layer as shown in Figure 4i. This quasi-bilayer structure offers advantages for electrical conductivity and temperature response. The bottom layer, which has a high concentration of PEDOT components, maintains its conduction pathways even when temperature and other factors fluctuate. Meanwhile, the PNIPAM-rich skin layer experiences changes in ion permeability due to morphological alterations associated with temperature variations, resulting in temperature response.

Figure 5a,b show the cyclic voltammograms of the 1:2 blend films with GOPS and PEGDE + GOPS crosslinkers, respectively. Both films exhibit capacitive behavior with volumetric capacitances (C^*) of 6.77 and 11.7 F cm^{-3}. Previous reports suggest that this capacitive behavior results from a process that replaces the holes extracted from the film by the metal contact with the injected cations from the electrolyte.[66] Figure 5c,d show the impedance spectra of these films (see Figures S9 and S10 (Supporting Information) for other blend ratios). We used a two-layer equivalent circuit to fit these curves. The model consists of a surface-segregated layer of PNIPAM and a bulk layer. The red lines in these panels are the fitting lines of the R_s-R_1//C_1-R_2//C_2

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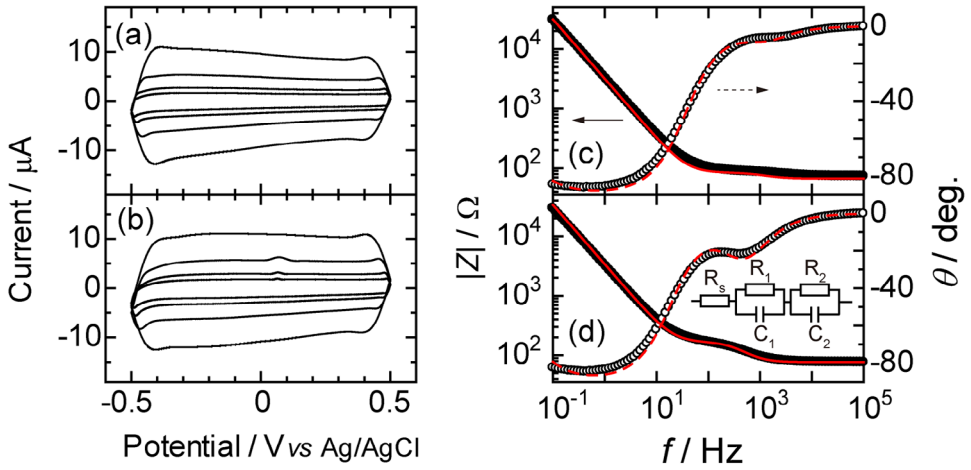


Figure 5. Electrochemical analysis of blend films: a,b) cyclic voltammograms and c,d) impedance spectra of PEDOT:PSS + PNIPAM(1:2) blend films crosslinked with (a,c) GOPS and (b,d) PEGDE + GOPS. (a,b) Cyclic voltammograms measured at 30, 50, 100, and 200 mV s^{-1}. (c,d) Red solid and broken lines show fitting lines by an equivalent circuit shown as the inset.

circuit with three components. The impedance spectra of the neat PEDOT: PSS films were fitted to a single element, $R_1//C_1$, as in our previous study.[14] This indicates that the additional $R_2//C_2$ component originates from the PNIPAM-rich phase in the film, which was indicated by the depth profile of XPS measurements above. The C_1^* values of the neat and blended films with GOPS single crosslinkers decrease linearly with the film composition, reflecting the PEDOT: PSS composition in the bulk layer. This is consistent with the conductivity behavior. However, the films with PEGDE + GOPS crosslinkers had higher C_1^* values than those with GOPS crosslinkers. We attribute this to the increased crystallinity of PEDOT in the bulk layer, as reported in previous experiments[76] and modeling studies.[66] The GIXD measurements support these capacitance results. Thus, the composition and crystallinity of PEDOT in the film determine the volumetric capacitance of the blended films.

Figure 6a shows the output and transfer curves of the OECT devices. These output curves indicate that the drain current decreased upon applying a positive gate voltage, which is consistent with depletion-mode operation arising from the de-doping of PEDOT upon the injection of cations from the electrolyte. Further, the transconductance, g_m versus V_G, curves (Figure 6b) show a typical bell shape for OECTs and a small hysteresis in the gate voltage scan for all compositions in the 1:2 blend. These

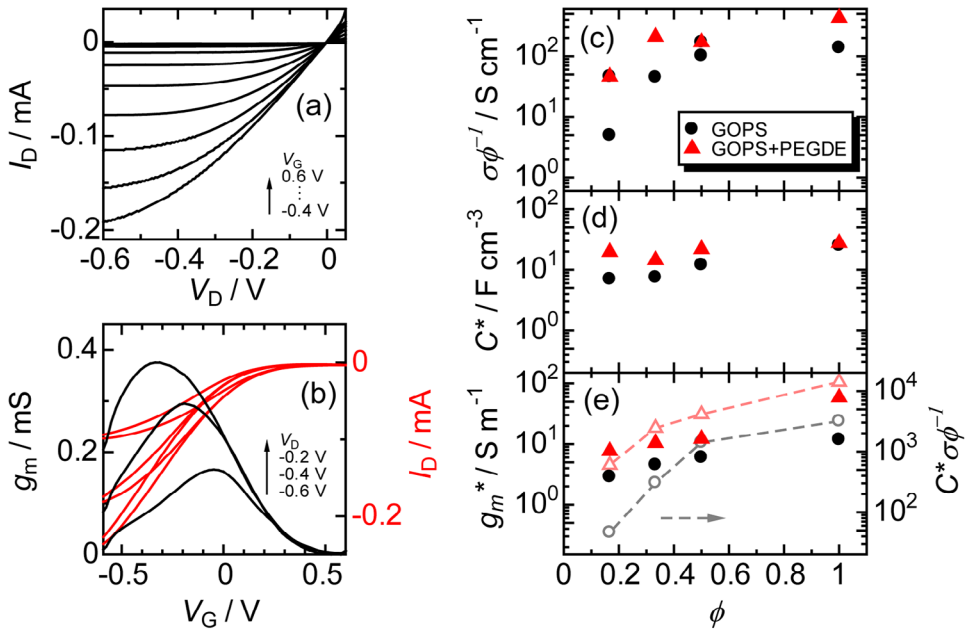


Figure 6. OECT device analysis of blend films: a) output and b) transfer curves (black: transconductance, red: drain current) of 1:2 blend films crosslinked with PEGDE + GOPS. (c,d,e) Transconductance (g_m), conductivity (σ) normalized by film composition (ϕ), and capacitance (C^*) plotted against the PEDOT:PSS contentϕ. Open symbols and broken lines indicate the product $C^*\sigma\phi^{-1}$.

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results indicate that the conduction path through the PEDOT in the channel was maintained for all blend ratios tested. For a more detailed analysis, we summarize the geometry-normalized transconductance ($g_m^* = g_m/WLt$, where W, L, and t are the width, length, and thickness of the channel, respectively) in Figure 6c. The introduction of PEGDE increased g_m^* for the neat and blended films, and the g_m^* of the neat films increased by approximately four times, likely because of the crystallinity enhancement. Previous studies have reported that the transconductance of OECT devices is proportional to the product of the hole (electron) mobility (μ) and capacity for ion injection C^*.[13] We can deconvolve the electronic mobility and the volume density of bulk charge storage based on the previous measurements; briefly, the mobility (μ) is related to the conductivity σ and carrier density (n) as $\mu = \sigma/n$. Figure 6d summarizes the parameter σ/ϕ as an indicator of electronic mobility μ assuming that the doping level of PEDOT is unchanged for all blend ratios. This assumption is justified by the abovementioned Raman and UV–vis spectra. The σ/ϕ varies over two orders of magnitude, reflecting the decrease in mobility due to the disruption of a percolative network of PEDOT that remains after subtracting the dilution effects of blending. This factor qualitatively follows the 3D random percolation behavior where the connection probability drops at around the critical threshold. Furthermore, the σ/ϕ of films with PEGDE + GOPS crosslinkers is higher than that of films with GOPS crosslinkers reflecting the crystallinity of the film. Using σ/ϕ and C^* obtained from cyclic voltammetry measurements (Figure 6e), the indicator of μC^* product $\sigma C^*/\phi$ is represented as open symbols and broken lines, respectively, in Figure 6c. Note that the y-axes of all panels in Figure 6c–e span three orders of magnitude. These trends almost follow the obtained g_m^* values, except for the 1:5 blend films. The origin of the deviation at a high PNIPAM content can be attributed to the conductance measurement conditions and OECT device performance in dry and wet conditions, respectively. Comparing the change in σ/ϕ and C^*, the electronic mobility contributed more to g_m^* than the capacitance for ion injection. In particular, the effect of the PEGDE crosslinkers on the OECT device performance is attributed to the enhancement of crystallinity, which affects the device performance through electronic mobility.

Next, we fabricated micropatterned devices to test device performance. The transconductance at $V_D = -0.5$ V of these devices was 664 ± 140 and 400 ± 26.2 mS for the blend films with GOPS and PEGDE + GOPS crosslinkers, respectively ($n = 5$, at 22 °C). In addition, we used temperature cycling across the T_c of PNIPAM to estimate its durability against volume expansion and shrinkage. **Figure** 7a illustrates the temperature dependence of the blend films with GOPS and PEGDE + GOPS crosslinkers. The device with the blend film crosslinked by GOPS showed an irreversible g_m response to temperature sweep between 20 and 40 °C; the g_m value decreased to 40% of the initial value after the heating process. The microscopic images captured after the temperature cycles (Figure 7b,c) show film cracks, indicating that the physical destruction of the conductive path caused an irreversible decrease in g_m (Figure 7d). In contrast, we did not observe such an irreversible behavior for the neat and blend films with the PEGDE + GOPS crosslinker (Figure 7e). Based on these observations, we concluded that the cracks were caused by the volume change of PNIPAM when the film crossed the critical temperature T_c. Notably, the decrease in transconductance in a lower temperature region at ≈20 °C compared to LCST temperature in DI water (32 °C) could be related to the shift in the critical temperature by local ionic strength in the channel layer due to PSS units.[77] Conversely, the blend film with the PEGDE + GOPS crosslinkers showed no cracking or delamination and irreversible behavior in g_m during the temperature cycle. The g_m increased upon heating and decreased upon cooling, having an average change of 4.0%/10 °C, whereas neat PEDOT:PSS showed only a small change in g_m (1.1%/10 °C), as previously reported.[78] Using the temperature dependent g_m values in Figure 7e, normalized response R was estimated as shown in Figure 7f to show the high sensitivity ≈20 °C. These results confirm that the reversible temperature response developed using the PEGDE crosslinker did not cause irreversible damage to the blended films.

Finally, we used electrochemical strain microscopy (ESM) to probe the swelling of the polymer due to ion injection into the film to elucidate ion penetration in the blend film.[48] **Figure** 8a–c show the ESM images of the blend film crosslinked with PEGDE + GOPS. Around the reduction onset potential (300 mV), the film swelled slightly by 1.33 ± 0.04 nm. With respect to the tip bias, the ESM amplitude exhibited considerable swelling because of ion/water injection into the film (Figure 8b,c). At the highest potential (+300 mV), the film swelled by 6.09 ± 0.03 nm, corresponding to the 6.7% swelling of the total film thickness. At a bias corresponding to the peak of g_m (0 V), the amplitude was 4.92 ± 0.11 nm with a significant dispersion compared to the other two cases. This indicates an inhomogeneous ion/water injection, particularly in the round regimes, which we ascribed to the PEDOT-rich domains, as shown in Figure 8b. The bias-dependent swelling behavior correlated with the decrease in the drain current of the transfer curve (blue line in Figure 8e), confirms that this ESM behavior reflects the de-doping and turn-off of the OECT devices discussed previously.

These findings indicate the factors to consider when designing a functional mixed conducting film based on a polymer blend system:[1] vertical phase separation and[2] appropriate selection of crosslinking agents. Vertical phase separation benefits a functionalized mixed conducting film, particularly with surface functionalities such as ion-selective operation or anti-fouling properties. The detailed mechanism of the vertical phase separation should be carefully investigated in a future study, yet the effect of molecular weight and hydrophobicity of PNIPAM during the annealing process are candidate driving forces for the self assembly.[79] Especially molecular weight might have a significant impact on surface segregation through chain-end motion. Mixed conductor films with graded compositions provide thermal and mechanical resilience, as in this study, and can also be an effective method for controlling biocompatibility and optical properties in the context of functionally graded materials (FGMs).[80] Further studies are required to understand the issues raised here, such as the control of the phase-separated structure and mechanical properties. On the other hand, applied research toward in vivo probing in terms of stability and reliability of device, circuit design, and integration is another important issue. These studies will pave the way for the facile preparation of functional biosensors or neuromorphic devices based on mixed conducting films in a polymer blend system.

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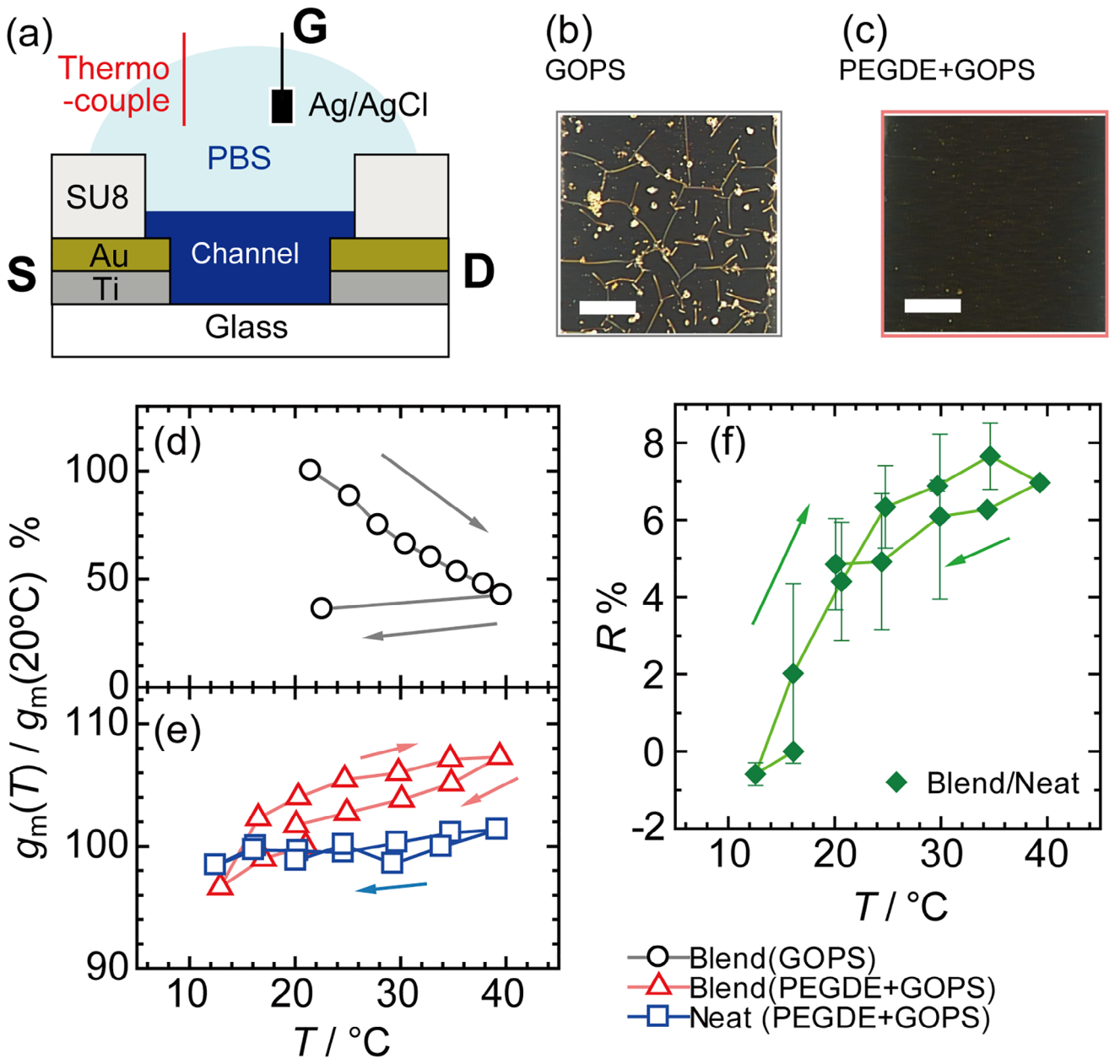


Figure 7. Temperature response of OECT devices: a) Illustration of measurement setup and a photograph of the channel region. b,c) Photograph of blend film crosslinked with (b) GOPS and (c) PEGDE + GOPS on gold electrode after temperature cycling (scale bar: 100 µm). d,e) Dependence of transconductance g_m on the temperature of the electrolyte 1:2 blend films of PEDOT:PSS + PNIPAM crosslinked with GOPS (black), PEGDE + GOPS (red), and PEDOT:PSS neat film crosslinked with PEGDE + GOPS. f) Normalized response of OECTs based on PEDOT:PSS + PNIPAM compared to those based on neat PEDOT:PSS over a full temperature cycle. Error bars indicate the standard deviation across two independent temperature scans performed on different devices.

3. Conclusion

We investigated OECTs based on blended films of PEDOT:PSS and the temperature-responsive polymer PNIPAM. The blended film had a segregated PNIPAM network on its surface. The flexible crosslinker PEGDE enhanced film stability and prevented crack formation and device degradation during temperature cycling. In addition, the addition of PEGDE increased the crystallinity, which improved both conductivity and volumetric capacitance. These findings are expected to contribute to advances in the emerging field of functional OECT devices for biosensors and other applications such as neuromorphic devices. Future studies on device optimization and mechanism analysis will pave the way for functional OECTs, including thermoresponsive and other biosensor OECTs with stimulus-responsive polymers.

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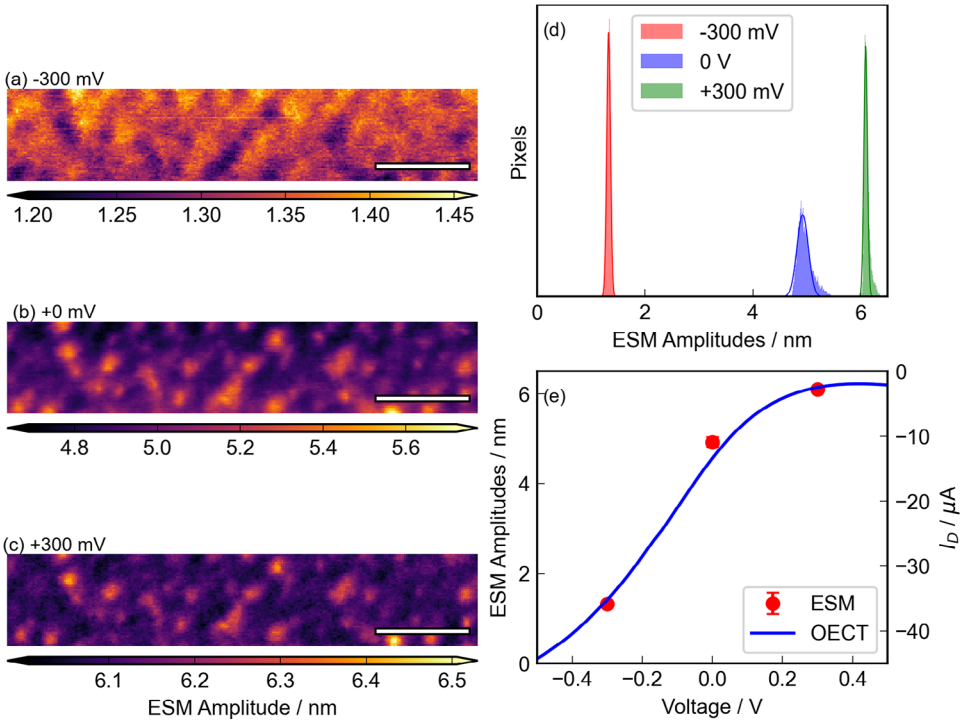


Figure 8. a–c) ESM Amplitude images of the 1:1 blend film of PEDOT:PSS + PNIPAM crosslinked with PEGDE + GOPS in 0.1 M KCl aqueous solution. ESM images were taken with the tip at different DC potentials over the threshold (as labeled). d) Histograms of the ESM amplitude of the same area under different bias conditions. e) Plot of ESM amplitude and drain current from transfer curve with $V_D = -0.1$ V. All scale bars correspond to 400 nm.

4. Experimental Section

PNIPAM Polymerization: The following compounds were weighed into a polymerization tube: 2.07 g (18.3 mmol) of NIPAM (Tokyo Kasei), 0.0105 g (0.0640 mmol) of 2,2′-azobis(isobutyronitrile) (AIBN) recrystallized twice with methanol (molar ratio to monomer: 3.50×10^{-3}), and 15 mL of anhydrous methanol (Wako Pure Chemicals).[81,82] The tubes were then frozen and degassed three times under vacuum. The polymerization reaction was initiated by placing the polymerization tube in an oil bath at 60 °C and continued overnight. After polymerization, the polymer was transferred to a round-bottom flask, and the solvent was removed using an evaporator. The polymer was dissolved in acetone, evaporated, and vacuum-dried. The polymer was then re-precipitated using acetone and hexane as solvents. The process was repeated three times, vacuum drying overnight to remove the monomers. The polymer was purified by dialysis using a cellulose tube (fractional molecular weight of ≈12000–14000, pore size of 5 nm) in DI water. The molecular weight of this polymer was estimated to be $M_w = 3.3 \times 10^4$; M_w/M_n was determined to be 1.6 by size-exclusion chromatography (SEC) measurements using polystyrene standards (Figure S1, Supporting Information). ^{1}H NMR (400 MHz, $CDCl_3$) δ = 0.9–1.2, 1.3–2.4, 3.8–4.2, 5.6–7.2 ppm (Figure S2, Supporting Information). The peak at ≈1–2 ppm indicates that this polymer contains both *m*- and *r*-diads.

Film Preparation: The premixed PEDOT:PSS solution was prepared using the literature method.[14] PEDOT:PSS (Clevios, PH 1000) and ethylene glycol (EG, FUJIFILM Wako Pure Chemical Co.) were used as the polar solvents, 4-dodecylbenzenesulfonic acid (DBSA, Tokyo Kasei) was used as the surfactant, and the mixture was sonicated for 15 min. The premixed mother solution (≈1.2 wt.% of PEDOT:PSS) was mixed with the 1.2wt.% PNIPAM aqueous solution in different ratios (PEDOT:PSS + PNIPAM = 1:0(neat), 1:1, 1:2, or 1:5 by weight basis). The crosslinkers GOPS (Tokyo Kasei) and PEGDE (Aldrich, average $M_n = 500$) at different molar ratios were prepared and added to the PEDOT:PSS + PNIPAM solution at the total volume ratio at 1%. The final solution was then sonicated for 15 min, spin-coated at 1000 rpm onto substrates, and annealed at 130 °C for 30 min. Each substrate was cleaned in an ultrasonic bath with acetone, isopropyl alcohol, and deionized (DI) water and treated with UV–ozone for 30 min before use. The obtained films were immersed overnight in DI water. For differential scanning calorimetry (DSC) measurements, solutions were cast on an aluminum dish and dried at 130 °C for 30 min to obtain cast films. DSC measurements were done using a calorimeter (DSC-60; Shimadzu Corp., Japan) in the presence of DI water.

Spectroscopic Analysis: Thin films were prepared on quartz substrates using the procedure described above. UV–vis measurements were performed using a spectrometer (V-670; JASCO). For the IR-p-polarized multiple-angle incidence resolution spectrometry (pMAIRS) and X-ray photoelectron spectroscopy (XPS) measurements, thin films were prepared on Si wafers using the procedure described above. IR-pMAIRS measurements were performed using a spectrometer (iS-50; Thermo Fisher Scientific) equipped with a sample rotation stage in dry air. XPS measurements were performed at the Nara Institute of Science and Technology using an XPS instrument (PHI5000VersaProbe II, ULVAC PHI). The measurements were conducted using a monochromatized AlKα X-ray source with an incident angle of 90° and an emission angle of 45°. The sputtering conditions were 10 kV and 20 nA for 10 min. Measurements were performed while etching the film surface with Ar-cluster ions.

Grazing Incidence X-Ray Diffraction (GIXD) Measurements: Thin films were prepared on Si wafers using the aforementioned procedure. GIXD measurements were performed using monochromated synchrotron radiation at the BL46XU beamline at SPring-8 with the approval of the Japan Synchrotron Radiation Research Institute (JASRI) (Proposal No. 2021B1953). The X-ray wavelength and incident angle were 0.1 nm and 0.12°, respectively. All measurements were performed under the ambient atmosphere.

Conductivity Measurements: Thin films were prepared on glass substrates using the procedure described above. Measurements were

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performed using the four-point probe method with a resistance meter (HIOKI 3541) connected to a four-point probe at a probe-to-probe distance of 5 mm (Loresta LSP; Mitsubishi Chemical Analytech).

Electrochemical Measurements: Electrochemical measurements were performed by depositing the blended film on an indium-tin-oxide (ITO) substrate. For electrochemical impedance spectroscopy (EIS) measurements, an LCR meter (ZM2376, NF Corp.) was used in the frequency range of 100 mHz to 100 kHz and a sine-wave AC voltage amplitude of 0.01 Vrms. Phosphate buffered saline (PBS; 0.0027 M KCl, 0.137 M NaCl) was used as the electrolyte with Ag/AgCl as the counter electrode and ITO as the working electrode. The cyclic voltammetry measurements were performed using a potentiostat analyzer (ECstat-301; EC Frontier Co., Ltd.) and the three-electrode method. PBS, Ag/AgCl, and Pt wire were used as the electrolyte, reference electrode, and counter electrode, respectively. The sweep rates were 20, 30, 50, 100, and 200 mV s^{-1}.

OECT Fabrication and Measurement: OECT substrates were prepared by depositing 10-nm Ti and 100-nm Au layers on glass substrates that had been pre-cleaned using the method described above. The substrate was spin-coated with SU-8 3005 and patterned in the channel region using a mask aligner (MA-10B, Mikasa). The self-assembled monolayer (SAM) treatment on electrodes was performed by immersing the substrate overnight in 0.5 mM α-thioglycerol (TG) (Tokyo Kasei) in dehydrated ethanol (FUJIFILM Wako Pure Chemical Co.) solution to deposit the SAM layer on the Au electrode.[83] The fabricated OECT devices (channel length (l) of ≈300 µm, channel width (w) of ≈1.0 mm) were analyzed using a semiconductor parameter analyzer (4155C; Keysight). The measurements were performed in a Faraday cage at a temperature-controlled stage using PBS as the electrolyte. The temperature of the device was controlled using a Peltier stage (HMC-11W-0100; Hayashi Repic) and monitored using a thermocouple immersed in an electrolyte solution.

Photoinduced Force Microscopy (PiFM) and Electrochemical Strain Microscopy (ESM) Observations: For PiFM, a Molecular Vista VistaScope coupled with a LaserTune QCL was used, with wavelength range ≈775 to 1850 cm^{-1}.[84] Measurements were acquired typically in sideband operation using gold-coated tips (Nanosensors PPP-NCHAu, f~300 kHz). Both the topography and PiFM images were processed in Python. The films were spin-coated onto an ITO substrate and imaged in ambient conditions. ESM measurements were performed on an Asylum Research Cypher-ES instrument using gold-coated contact mode tips (BudgetSensors CONTGB-G).[84] The films were spin-coated on an ITO substrate and immersed in 0.1 M KCl electrolyte for the ESM observation. A DC bias was applied to the electrolyte between the tip and sample and electrically grounded, whereas a drive bias was applied to the tip, using dual-amplitude resonance tracking as described previously.[44]

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

SY designed and conceived the experiments with help from SK and MM. SK fabricated OECT devices. SK and SY characterized the devices. RM and AK performed GIXD measurements, data analysis, and interpretation. SY, RG, and DSG performed PiFM and ESM measurements, data analysis, and interpretation. The manuscript was written with contributions from all authors. All authors have given approval to the final version of the manuscript.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

crosslinker, mixed-conducting polymer, organic electrochemical transistor, surface segregation, thermoresponsive polymer

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